

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

*in the name of Allah, the most beneficent,
the most merciful*





Eukaryotic cells , extracellular fluid and Water Balance

By

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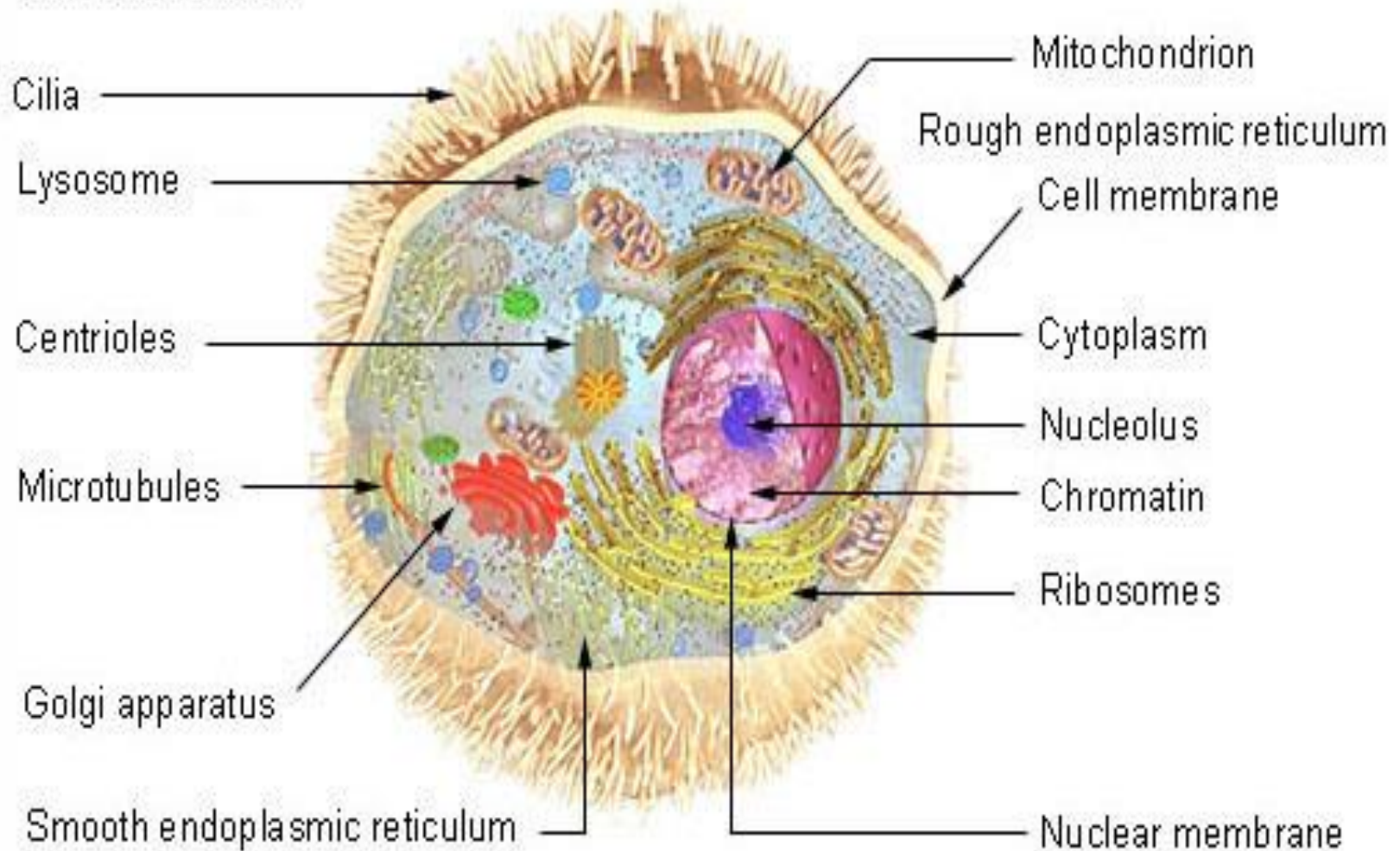
**Prof. of Medical Biochemistry &
Molecular Biology**



By the end of the lecture the student will be able to:

1. Describe the biochemical structure of cells.
2. List the two main types of cells
3. Illustrate the sources and different types of Stem cells
4. Define apoptosis and neoplasia.
5. Enumerate cell membrane functions.
6. Define extra and intracellular fluids

Cell Structure



- Cells consist of a collection of simple and complex organic molecules, inorganic molecules, and ions. There are also organized clusters of these called supramolecular complexes. The majority of biomolecules are composed of only 6 chemical elements - C, H, O, N, S and P.

- ❑ Prokaryotes - Small, single cell organisms lacking a membrane-bound nucleus and other subcellular compartments. Bacteria and blue-green algae are examples.
- ❑ Eukaryotes - Cell or organism with membrane-bound nucleus and other well-developed subcellular compartments (organelles). Yeast, fungi, protozoans, plants and animals are eukaryotes.

Prokaryotes

1 - Has ill defined nuclear membranes

2 - No or less subcellular organelles as mitochondria, Golgi apparatus

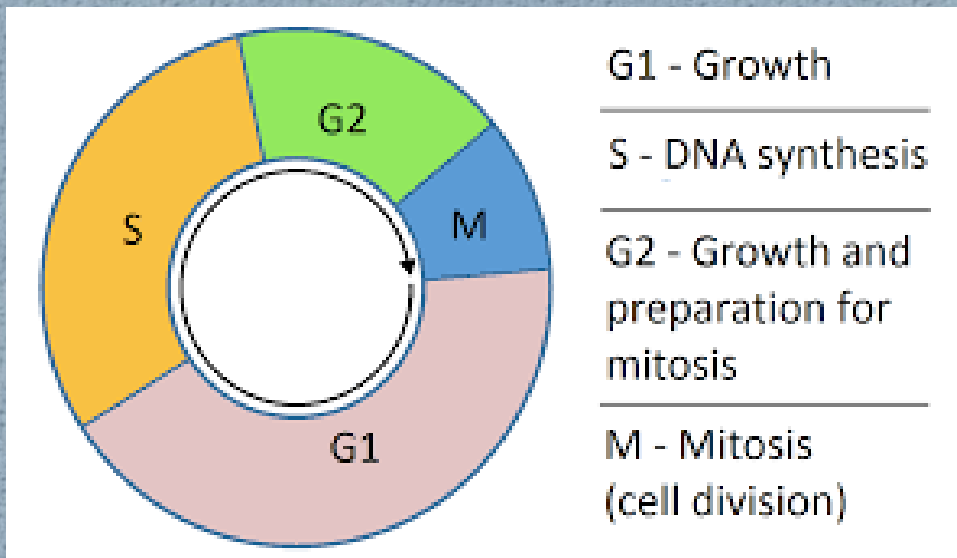
Eukaryotes

1- Well defined nuclear membranes

2- Rich in subcellular organelles.

CELL CYCLE

Cells proliferate as they pass through the cell cycle and undergo mitosis. Cells destroy themselves when they are no longer needed by apoptosis. Two groups of proteins, called cyclins and cyclin-dependent kinases (Cdks), are responsible for the progress of the cell through the various checkpoints.



Definition of Apoptosis it is genetically regulated form of cell death that permits the safe disposal of cells. It is a programmed controlled process of cellular self-destruction that requires certain protein releases and specific cellular signals

Neoplasia: is an abnormal accumulation of cells that occurs because of an imbalance between cellular proliferation and cellular attrition.

STEM CELLS

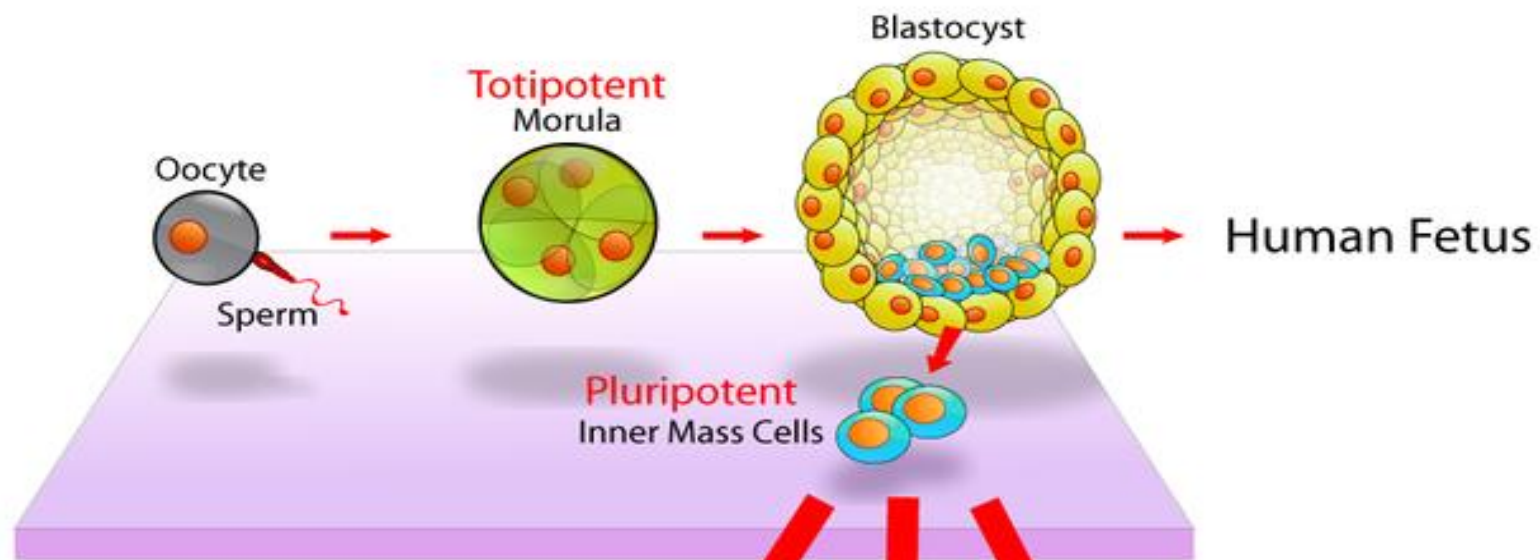
Stem cells: They are biological cells found in all multicellular organisms. They are responsible for the development and the regeneration of the tissue and organ systems. Stem cells are defined as clonogenic cells capable of both self-renewal and multi-lineage differentiation.

Sources of stem Cells

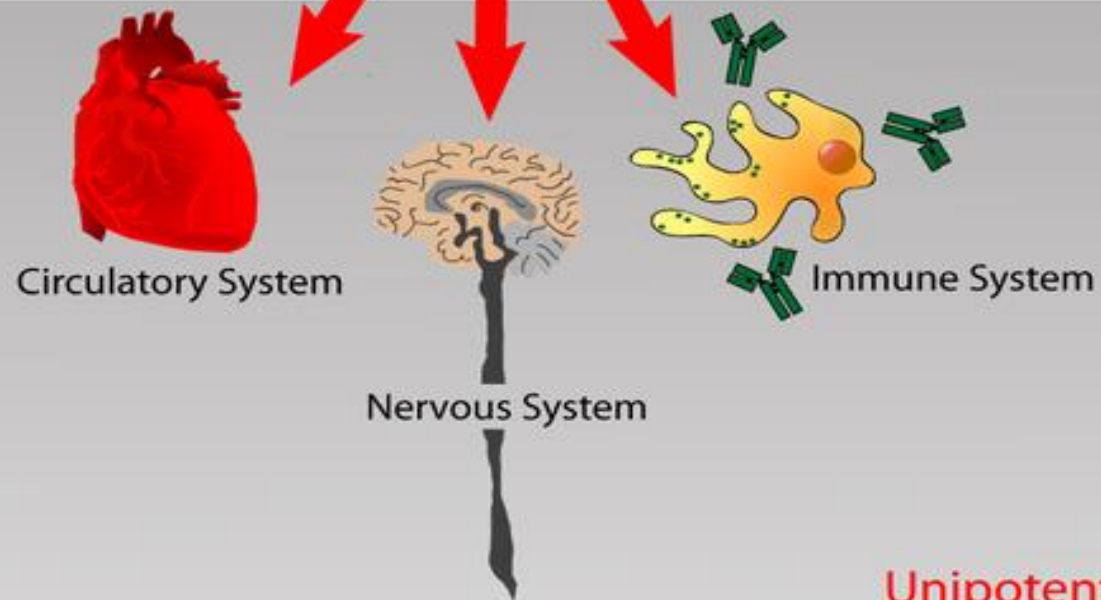
1. Embryonic stem cells (ESCs): derived from early human embryos.
2. Fetal stem cells (FSCs): from placenta, membranes, amniotic fluid or fetal tissues.
3. Adult stem cell (ASCs): accessible sources of autologous stem cells in humans:
 - 3.1. Bone marrow:
 - 3.2. Adipose tissue:
 - 3.3. Blood through pheresis wherein blood is drawn from the donor[similar to blood donation] passed through machine that extracts the stem cells and returns the other portions of blood.

Types of stem cells:

- 1. Totipotent:** Have the ability to become any thing from a complete human being to any type of tissue cells.
- 2.Pluripotent:** derived from the inner mass of blastocyste and can produce number of tissue but not have the ability to produce an entire human being.
- 3...Multipotent:** more specialized stem cells are found in adult and children e.g hemopiotic stem cells that differentiate to red blood cells, white blood cells, platelets.
- 4.Unipotent:** produce only one cell type with probability of self-renewal.



Examples:

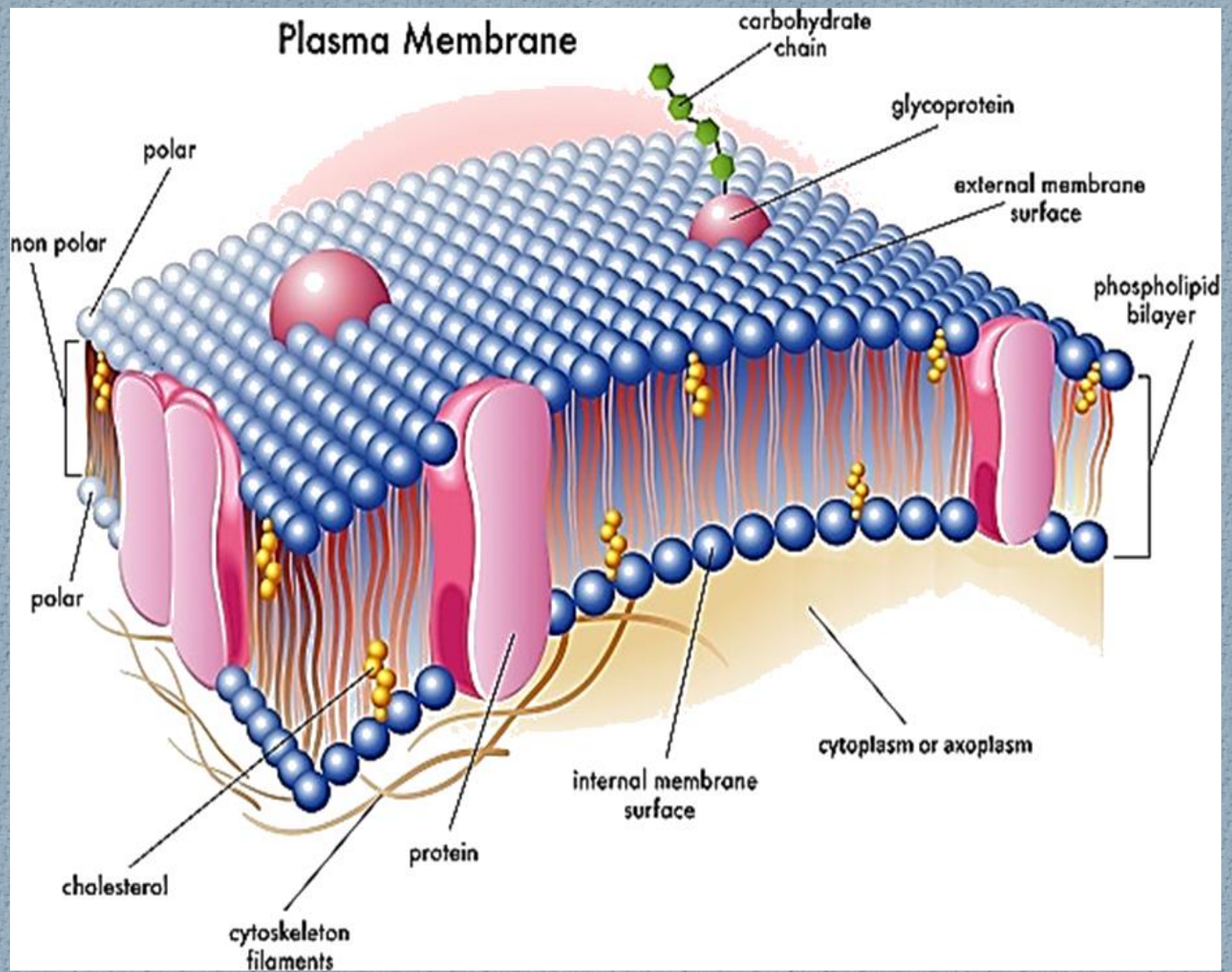


Unipotent

Overview about cell chemical composition

- Hydrogen, oxygen, nitrogen, carbon, sulfur, and phosphorus normally makeup the main mass of living cells.
- Ninety-nine percent of the molecules inside living cells are water molecules.
- Cells normally contain more protein than DNA.
- Non-essential lipids can be generated from acetyl-CoA.
- Certain amino acids and unsaturated fatty acids, various inorganic elements (minerals) are essential.
- Most of diseases in animals are manifestations of abnormalities in biomolecules, chemical reactions, or biochemical pathways.

Plasma Membrane



Membranes are thin envelopes that define the volume of cells and define cellular organelles. They serve a number of essential cellular functions:

- 1- Membranes constitute the boundaries of cells and intracellular organelles. They provide a surface where many important biological reactions, processes occur and interaction between cells.
- 2- Membranes have proteins that mediate and regulate the transport of metabolites, macromolecules and ions.
- 3- Hormones and many other biological signal molecules and regulatory agents exert their effects via interaction with membranes which have specific receptors.

4- Photosynthesis, electron transport, oxidative phosphorylation, muscle contraction and electrical activity all depend on membranes and membrane proteins.

5- Plasma membranes exchange material with extracellular environment by exocytosis and endocytosis.

6- Intracellular membranes form many of the morphologically distinguishable organelles e.g mitochondria, Golgi complex, endoplasmic reticulum, lysosomes and nuclear membranes.

Water Balance

WATER accounts for about 50 – 60% of an average person's weight.

- Fat tissue has lower percentage of water and women tend to have more fat, so the percentage of water in the average woman is lower (**50 to 55%**) than it is in the average man (**60%**).
- The percentage of water is also lower in **older** and **obese people**.

- A 70 kilogram man has about 45 liters of water in
- his body:

1. Intracellular Fluid ($\frac{2}{3}$ of body water) = 30 L

2. Extracellular Fluid ($\frac{1}{3}$ of body water) = 15 L

- Interstitial Fluid ($\frac{4}{5}$ of extracellular fluid) = 12 L
- Plasma ($\frac{1}{5}$ of extracellular fluid) = 3 L

How the body can regulates water?

1. The body obtains water primarily by **absorbing it from the digestive tract** in addition to the **small amount of water produced during metabolism**.
2. **Water intake must balance water loss** to maintain water balance and -- to protect against dehydration, the development of kidney stones, and other medical problems .
3. **Healthy adults should drink at least 1.5 - 2.0 liters/day.**
4. **Drinking too much is better than drinking too little**, because excreting excess water is much easier for the body than conserving water.
5. **Antidiuretic hormone (ADH):** secreted from the posterior lobe of pituitary gland maintain water at equilibrium through regulation of water absorption from the distal convoluted tubules of the kidney according to body needs.
5. The body regulates the amount of water in each of intracellular and extracellular compartments by **moving water between these areas** to keep **(Osmolality)** the amount in each area relatively constant .

Reference

- 1- Harpers illustrated Biochemistry
- 2- Lippincott illustrated reviews integrated systems

